6G Weekly Survey — Nov 09 - Nov 15, 2026

15 items across 5 topics, ranked by importance.

6G Demos & Trials

Sensing-Based Communication: A Non-Orthogonal Passive Reflection Framework for ISAC

Xin Xiong, Mathew Salas, Sihua Shao, Chen Chen, et al. · arXiv cs.IT · 2026-11-06

★★★★■ Strong contribution addressing a fundamental ISAC challenge with validated theory and experimental proof-of-concept, though optical testbed limits immediate RF applicability and commercial readiness remains unclear.

This paper introduces a non-orthogonal passive ISAC framework called Sensing-based Communication, where passive tags parasitically modulate incident downlink signals using simple on-off keying to enable uplink without consuming additional spectrum or reducing downlink rates. The approach unifies the reflective sensing channel and uplink communication channel, employing joint cross-correlation and successive interference cancellation at the receiver to suppress multi-user interference. Validated experimentally in an optical retro-reflective testbed with OFDM waveforms, the theory accurately predicts SINR scaling with bandwidth and correlation time (R^2=0.97 experimental, R^2=0.94 simulation).

Contribution: Novel wave-agnostic passive ISAC architecture that achieves simultaneous sensing and full-duplex communication by unifying sensing and uplink channels through parasitic OOK modulation of downlink signals, requiring no additional spectrum.

Relevance to your work: Directly relevant to your ISAC research interests, offering a passive, spectrum-efficient architecture for joint sensing-communication that could inform backscatter/ambient designs in 6G and extends ISAC beyond traditional active radar paradigms.

Beyond Feasibility: Efficient and Robust On-Device Localization with UWB Concurrent Transmissions

Enrico Soprana, Michael Stocker, Davide Vecchia, Matteo Trobinger, et al. · arXiv eess.SP · 2026-11-06

★★★■■ Solid incremental work addressing practical deployment barriers for UWB localization, relevant for ISAC and NTN use cases requiring efficient on-device positioning, though not fundamentally changing 6G physical layer design.

This paper advances UWB concurrent transmissions (CTX) for tag self-localization by enabling efficient on-device execution on resource-constrained MCUs and adding robustness to interference via IEEE 802.15.4z features. The redesigned CTX techniques allow a tag to range to multiple anchors with a single frame, achieving 44% runtime reduction and 85% energy savings versus traditional sequential approaches. Real-world testbed validation demonstrates practical viability with dynamic anchor selection and interference mitigation.

Contribution: First practical implementation of CTX-based UWB localization running entirely on resource-constrained tags with interference robustness through IEEE 802.15.4z, moving beyond proof-of-concept to deployable systems.

Relevance to your work: Tangential—focuses on UWB ranging/localization implementation rather than AI-native PHY or advanced sensing-communication integration, though the interference mitigation and efficiency aspects could inform ISAC beam management strategies.

Stable probabilistic emulation of partially observed dynamical systems

Mario Massy, Sara Shamekh, Fabrizio Falasca, Matthieu Blanke · arXiv cs.LG · 2026-11-06

★★★■■ Solid theoretical contribution to stable generative modeling with rigorous stability proofs, though applied to climate/atmospheric systems rather than wireless channels.

The paper introduces forced-dissipative stochastic interpolants for learning stable emulators of partially observed dynamical systems with unresolved fast variables. The architecture incorporates structural

constraints from turbulent dynamics to guarantee long-term stability while capturing complex probabilistic behaviors. Evaluated on the two-timescale Lorenz-96 system, the method demonstrates geometric ergodicity, stability under external forcing, and accurate prediction of forced responses in both mean and variance.

Contribution: Provides provable geometric ergodicity and stability guarantees for stochastic generative emulators through Foster-Lyapunov drift conditions, addressing the critical challenge of long-term stability in data-driven reduced-order models.

Relevance to your work: Tangential: the forced-dissipative framework could potentially inform stable generative channel models (e.g., GAN-based temporal channel evolution), but the atmospheric physics focus is distant from wireless physical layer concerns.

Encoding Memory in World Models with Test-Time Model-Based Reinforcement Learning

Maxime Burchi, Fayaz Ali Dharejo, Radu Timofte · arXiv cs.LG · 2026-11-06

★★■■■ Competent RL work with solid empirical results, but addresses general sequential decision-making rather than wireless-specific challenges; methodology does not translate to wireless PHY or protocol optimization.

This paper addresses long-term memory limitations in model-based reinforcement learning by introducing Test-Time MBRL, which encodes episodic observations into world model parameters via gradient descent at inference time rather than relying solely on architectural context memory. The approach learns episode-specific world models, achieving state-of-the-art on Memory Maze benchmarks while maintaining performance on standard RL tasks. Validation spans simplified 2D environments through full 3D domains and non-memory control tasks.

Contribution: Shifts from architectural design to test-time parameter adaptation for encoding long-term memory in world models, enabling episode-specific learning that outperforms context-based approaches.

Relevance to your work: Tangential—while world models and online learning appear in wireless contexts (environment prediction, beam tracking), this work targets game-playing domains with no connection to air interface design, channel modeling, or communication system architecture.

Robust Deep Prototype Classifiers via Lipschitz Training

Adia Khalid Khan, Lukas Gehring, Benjamin Paaßen · arXiv cs.LG · 2026-11-06

★★■■■ While technically sound for computer vision robustness, this work addresses general ML adversarial training rather than wireless-specific challenges and has no clear connection to physical-layer learning, channel models, or communication system design.

This paper addresses robustness in deep prototype-based classifiers by constraining the Lipschitz constant of embedding networks. The authors prove that if an embedding network is locally L-Lipschitz and the prototype classifier is ε-robust in embedding space, the full model is ε/L-robust in input space. Experiments on MNIST, CIFAR-10, and Imagenette-160 show that combining Lipschitz-constrained embeddings with prototype-based heads (especially GLVQ) provides superior robustness against FGS, PGD, and C&W attacks compared to standard embeddings.

Contribution: The paper provides a theoretical bound linking embedding network Lipschitz continuity to end-to-end classifier robustness, and demonstrates a practical Lipschitz training scheme that empirically improves adversarial robustness of prototype classifiers on image data.

Relevance to your work: tangential

ICare-DT: a digital twin for continuous forecasting of intensive care patients

Yanke Li, Yahang Qi, Yumi Kim, Diego Paez-Granados · arXiv cs.LG · 2026-11-06

★■■■■ While methodologically sound for medical ML, this work is entirely healthcare-focused with no connection to wireless physical layer, communication systems, or network optimization.

ICare-DT is a Transformer-based digital twin for ICU patients that updates forecasts hourly from new measurements without retraining, improving vital-sign prediction accuracy by 0.025–0.153 AUROC as more patient data becomes available. The authors demonstrate forecast improvement on eICU-CRD and MIMIC-IV datasets and introduce a benchmark for evaluating treatment-response prediction using simulated interventions injected into real patient records. Standard learners failed to predict responses outside their training support, but informed models achieved R^2 of 0.30–0.40 when given the generative structure.

Contribution: First demonstration that Transformer-based patient state tracking can improve forecasts incrementally from routine ICU measurements without retraining, plus a rigorous benchmark exposing fundamental limitations in causal treatment-response prediction from observational data.

Relevance to your work: Tangential—digital twin methodology may inform future 6G network twins, but the ICU application domain, time-series forecasting problem, and evaluation methods do not translate to wireless research priorities.

LACART: Lookahead-Augmented Collision Avoidance and Reference Tracking Strategies for Safety-Critical Robots

Pedro Lopes, Amit Shivam, Luis Tiago Paiva · arXiv eess.SY · 2026-11-06

★■■■■ This is ground-robot control engineering with no connection to wireless physical layer, channel modeling, or communication system design for 6G.

This paper develops LACART, a collision-avoidance system for autonomous racing robots that combines lookahead control barrier functions (CBFs) with control Lyapunov functions (CLFs) and quadratic programming to modify vehicle commands. Hardware validation demonstrates 25 Hz control on a 1:43-scale vehicle using camera-based estimation and wireless actuation. Simulation shows comparable performance to existing methods with slightly lower median computation times (0.31 vs 0.33 ms per step).

Contribution: The method unifies lookahead-based CBFs with track boundary constraints in a single QP optimization layer and validates real-time safety intervention on physical hardware at 25 Hz.

Relevance to your work: tangential

ReplayAudit: Auditing Outsourced LLM Fine-Tuning via Sealed Records and Selective Replay

Yihan Xia, Wenxin Xu, Taotao Wang · arXiv cs.LG · 2026-11-06

★■■■■ This work addresses LLM training verification and cloud computing trust, which has no connection to wireless physical layer, air interface design, channel modeling, or communication system optimization relevant to 6G research.

This paper addresses verification of outsourced large language model fine-tuning by proposing ReplayAudit, a system that seals training records and selectively replays computation intervals from checkpoints. The method uses training telemetry (loss changes, parameter-update norms) to guide stratified sampling, focusing verification effort on suspicious portions of training rather than uniform sampling. Experiments on LoRA fine-tuning of billion-parameter models show 49-67% reduction in audit wall-time for certain anomaly types, though evasion of high-anomaly strata can increase costs by 38%.

Contribution: The novel contribution is a telemetry-guided stratified sampling approach for verifying outsourced LLM training that achieves deterministic replay from sealed checkpoints while reducing audit costs compared to uniform sampling when anomalies correlate with training metrics.

Relevance to your work: tangential

AI-native air interface

AI-Native Task-Oriented Semantic Communications over a Relay: Benefits and Vulnerabilities

Yalin E. Sagduyu, Tugba Erpek, Aylin Yener · arXiv cs.IT · 2026-11-05

★★★★■ Addresses a critical but underexplored security dimension of AI-native air interfaces—insider threats in semantic communication—with practical relevance as semantic/task-oriented approaches mature toward 6G deployment.
This paper studies semantic communication through a relay node, designing a joint end-to-end encoder-relay-decoder system for image classification and reconstruction tasks. Relay-assisted schemes outperform direct transmission especially at low SNR, but the authors identify a critical vulnerability where a Byzantine relay can inject learned perturbations into intermediate representations to degrade semantic inference accuracy. The work quantifies a fundamental trade-off between performance gains and security risks in relay-assisted semantic communication.

Contribution: First systematic study of Byzantine attacks on semantic communication relays, demonstrating that adversarial perturbations applied to learned intermediate representations under channel effects can significantly compromise task performance.

Relevance to your work: Directly relevant to your semantic and task-oriented communication research, highlighting security considerations that must be integrated into AI-native PHY designs, especially for relay/multi-hop scenarios in Beyond-5G architectures.

ISAC (sensing + communication)

Physical Sensing Verification for Cognitive ISAC against UAV Spoofing via Digital Twins

Seyed Bagher Hashemi Natanzi, Bo Tang · arXiv eess.SP · 2026-11-05

★★★★■ Addresses a critical security gap in cognitive ISAC systems with a novel physics-based approach, credible methodology (pre-registered experiments), and practical relevance as UAV detection becomes operationally deployed in 5G/6G.
The paper addresses spoofing attacks in cognitive ISAC systems where an adversarial relay replays base station signals to create phantom UAV targets. It proposes a wall-based physical verification method using a digital twin to predict environmental reflections that a relay cannot authentically replicate. Ray-tracing evaluation demonstrates perfect separation (AUC 1.000) between genuine targets and relay attacks using passive environmental echo analysis.

Contribution: First physical-layer anti-spoofing mechanism for ISAC that exploits the impossibility of a relay faking environmental multipath geometry, validated through pre-registered digital twin simulations with provable detection bounds under specified relay conditions.

Relevance to your work: Directly relevant to ISAC research: demonstrates how sensing infrastructure can defend itself using environmental knowledge and digital twins, pointing toward intelligent physical-layer security mechanisms for 6G ISAC systems where sensing outputs drive automated responses.

Sparse Array-Inspired Frame Structure Design for 5G NR ISAC

Haisu Wu, Hong Ren, Cunhua Pan, Yuqing Chen, et al. · arXiv eess.SP · 2026-11-06

★★★★■ Directly addresses a critical resource allocation problem in practical ISAC deployment for 5G NR, with a mathematically principled approach that bridges sparse array theory and standardized frame structures.
This paper addresses the time-frequency resource conflict between sensing and communication in ISAC by proposing a sparse sensing frame structure for 5G NR that maps 2D sparse antenna array geometries onto the resource grid. By leveraging difference coarray properties to eliminate range-velocity ambiguity, the authors cast sensing signals into higher-order tensors and develop a binary-tree-based tensor train decomposition (BT-TTD) algorithm for parameter estimation. Simulations demonstrate competitive estimation accuracy with reduced complexity compared to dense resource allocation.

Contribution: Novel mapping of sparse array design principles to time-frequency resource allocation in 5G NR ISAC, enabling ambiguity-free sensing with reduced resource occupancy through tensor-based signal processing.

Relevance to your work: Highly relevant to your ISAC research interests, as it provides a concrete framework for efficient sensing resource design compatible with existing 5G standards while advancing

tensor-based signal processing methods for joint radar-communication systems.

Pinching Antenna Systems (PASS) for Air-Ground Integrated Sensing and Communications (ISAC)

Chaoran Chen, Haochen Li, Jincheng Zhu, Zhiwen Pan, et al. · arXiv eess.SP · 2026-11-06

★★★■■ Solid incremental advance in ISAC system design combining movable antennas and UAVs with rigorous optimization, though limited to single-target scenarios and simulated validation only.

This paper addresses multi-user ISAC with a novel ground-based pinching antenna system (PASS) transmitting to ground users while a UAV collects target echoes. The authors jointly optimize communication/sensing beamforming, movable antenna positions on the ground, and UAV trajectory to minimize weighted Cramér-Rao bound for target localization. A penalty dual decomposition algorithm with provable KKT-point convergence achieves lower CRB than fixed MIMO and PASS baselines.

Contribution: First work to combine movable/fluid antenna arrays at the ground station with UAV-based echo collection for air-ground ISAC, with joint optimization of beamforming and physical geometry backed by convergence guarantees.

Relevance to your work: Directly relevant to your ISAC research interests; the movable antenna concept could inform reconfigurable surfaces or advanced beam management for integrated sensing in NTN/UAV scenarios, though the physical feasibility of continuous antenna repositioning remains unclear.

NTN / satellite

Data-Driven Spectrum Access Technologies for 6G

Foad Sohrabi, Harish Viswanathan, Sivarama Venkatesan, Jinfeng Du, et al. · arXiv cs.IT · 2026-11-02

★★★★■ Addresses a critical 6G deployment barrier (spectrum scarcity) with credible industrial authorship and practical ML-based solutions aligned with regulatory realities and NTN integration trends.

This paper addresses spectrum scarcity for 6G by proposing four AI-driven co-existence techniques to share spectrum below 15 GHz with incumbent services (satellite, radar). Methods include: protection zone management for fixed receivers, horizon-restricted power control for satellite protection, fast PRB blanking via spectrum sensing, and base-station occupancy detection for dynamic sharing with NTN systems. All leverage AI and massive MIMO to enable efficient sharing while preserving cellular performance.

Contribution: First comprehensive framework combining AI and massive MIMO for practical spectrum co-existence across multiple incumbent scenarios (fixed, satellite, NTN), with fast-timescale adaptation mechanisms.

Relevance to your work: Directly relevant: combines AI/ML with NTN co-existence and massive MIMO beamforming—three of your core interests—and proposes physical-layer techniques applicable to beyond-5G spectrum sharing challenges.

6G Physical Layer

Payload-Blind Bistatic Delay-Doppler Sensing with Phase-Injected Chirps in CE-OFDM

Gustavo P. Gonçalves, Henrique L. Senger, Bruno S. Chang, Giuseppe Thadeu Freitas de Abreu · arXiv eess.SP · 2026-11-06

★★★★■ Solves a fundamental tension in ISAC waveform design—achieving sensing performance without sacrificing PAPR or communication efficiency—using a practical method applicable to next-generation constant-envelope air interfaces.

The paper proposes injecting a known chirp (Zadoff-Chu sequence) into the phase of constant-envelope OFDM (CE-OFDM) to enable bistatic delay-Doppler sensing at a receive-only node without dedicated pilots or disrupting the constant envelope. The method exploits the ensemble mean of the payload as an implicit

pilot, using Doppler-aligned coherent integration across frames to suppress payload interference. Simulations show only 2.8 dB loss versus payload-aware matched filtering while eliminating 53% frame overhead of dedicated pilots.

Contribution: First demonstration of payload-blind bistatic ISAC using phase-injected pilots in CE-OFDM that preserves constant envelope, avoids dedicated sensing resources, and achieves near-optimal detection through Doppler-aligned averaging.

Relevance to your work: Directly advances AI-native ISAC physical layer design by enabling transparent sensing in CE-OFDM, a candidate 6G waveform that addresses PAPR constraints while supporting joint communication and sensing with minimal overhead.

A Novel BVH-accelerated Dynamic Ray-Tracing Channel Model for Vehicular Communication Systems

Songjiang Yang, Cheng-Xiang Wang, Yinghua Wang, Chen Wang, et al. · arXiv eess.SP · 2026-11-05

★★★■■ Solid incremental contribution addressing computational efficiency in ray-tracing for high-mobility scenarios, with practical validation for 6G delay-Doppler waveforms, though the core innovation is computational rather than fundamentally advancing channel modeling theory.

This paper proposes BVH-accelerated dynamic ray-tracing (BVH-DRT) that replaces time-driven updates with spatial-driven updates based on geometric coherence distance, reducing computational complexity in high-mobility vehicular channel modeling. The method clusters scatterers using bounding volume hierarchy preprocessing and autonomously adjusts ray update intervals based on transceiver kinematics. Validation in V2V scenarios shows spatial-temporal consistency with measurements while improving efficiency, and BER evaluations demonstrate performance of delay-Doppler waveforms (OTFS, ODDM) under high-dynamic channels.

Contribution: The spatial-driven update mechanism using BVH clustering and geometric coherence distance represents a novel approach to reduce ray-tracing complexity in dynamic channels while maintaining accuracy for non-stationary vehicular scenarios.

Relevance to your work: Directly relevant to your 6G waveform interests (OTFS) and channel modeling work, providing efficient simulation tools for evaluating AI-native PHY designs in non-stationary vehicular and potentially NTN high-Doppler environments.